

Co-optimization of Rooftop Solar and Storage under NEM 3.0

Model Specification (Step 9b)

Ana Santasheva

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1 Overview

Step 9b is a linear program that jointly chooses the rooftop solar array size, the battery energy and power capacity, and the hour-by-hour dispatch of solar, battery, and grid, so as to minimize a household's total annual cost. Total annual cost is the sum of annualized capital cost and the net annual energy bill (imports priced at the retail time-of-use rate, exports credited at the NEM 3.0 Avoided Cost Calculator rate). The program is solved at full 8760-hour resolution, or on a reduced set of 288 representative time slices (12 months $\times$ 24 hours) weighted by days per month. All variables are continuous; there are no integer variables.

2 Nomenclature

Sets and indices

Symbol	Meaning
h	index of an hour (or representative time slice)
$\mathcal{H}$	set of hours in the analysis year: 8760, or 288 representative slices

Decision variables (all ≥ 0 , continuous)

Symbol	Meaning	Units
PV	installed solar (photovoltaic) capacity	kW
B_E	battery usable energy capacity	kWh
B_P	battery power (inverter) capacity	kW
$x_h^{pv \rightarrow l}$	solar energy sent to serve load in hour h	kWh
$x_h^{pv \rightarrow b}$	solar energy sent to charge the battery	kWh
$x_h^{pv \rightarrow g}$	solar energy exported to the grid	kWh
$x_h^{b \rightarrow l}$	battery energy discharged to serve load	kWh
$x_h^{b \rightarrow g}$	battery energy exported to the grid	kWh
$x_h^{g \rightarrow l}$	grid energy imported to serve load	kWh
$x_h^{g \rightarrow b}$	grid energy imported to charge the battery	kWh
soc_h	battery state of charge at the start of hour h	kWh

Parameters

Symbol	Meaning	Units
L_h	household electricity load in hour h	kWh
G_h	solar generation per kW installed in hour h (AC yield)	kWh/kW
p_h^{imp}	retail import price in hour h (time-of-use)	\$/kWh
p_h^{exp}	export credit in hour h (NEM 3.0 ACC)	\$/kWh
w_h	weight of hour/slice h (days in month for 288 mode; 1 for 8760)	—
c_{pv}	solar installed capital cost	\$/kW
c_b	battery energy installed capital cost	\$/kWh
c_{bp}	battery power installed capital cost (default 0)	\$/kW
c_{deg}	battery degradation cost per kWh discharged (default 0)	\$/kWh
r	real discount rate (default 0.07)	—
N	analysis horizon and solar service life (25)	yr
n_{batt}	battery service life (15)	yr
η_{ch}, η_{dis}	battery charge / discharge efficiency, each $\sqrt{0.96}$	—
$\underline{s}, \bar{s}$	minimum / maximum state-of-charge fraction (0.20, 0.90)	—
α_{pv}, α_b	solar / battery capital-annualization factors (see §4)	1/yr

3 Optimization problem

Objective. Minimize total annual cost: annualized capital cost, plus the net annual energy bill (imports minus export credits), plus battery degradation cost.

$$\begin{aligned}
\min \quad & PV c_{pv} \alpha_{pv} + B_E c_b \alpha_b + B_P c_{bp} \alpha_b \\
& + \sum_{h \in \mathcal{H}} w_h \left[(x_h^{g \rightarrow l} + x_h^{g \rightarrow b}) p_h^{imp} - (x_h^{pv \rightarrow g} + x_h^{b \rightarrow g}) p_h^{exp} \right] \\
& + \sum_{h \in \mathcal{H}} w_h c_{deg} (x_h^{b \rightarrow l} + x_h^{b \rightarrow g})
\end{aligned} \tag{1}$$

Constraints. For every hour $h \in \mathcal{H}$:

$$x_h^{pv \rightarrow l} + x_h^{pv \rightarrow b} + x_h^{pv \rightarrow g} \leq PV \cdot G_h \quad (\text{solar availability}) \tag{2}$$

$$x_h^{pv \rightarrow l} + x_h^{b \rightarrow l} + x_h^{g \rightarrow l} = L_h \quad (\text{load balance}) \tag{3}$$

$$x_h^{pv \rightarrow b} + x_h^{g \rightarrow b} \leq B_P \quad (\text{charge power}) \tag{4}$$

$$x_h^{b \rightarrow l} + x_h^{b \rightarrow g} \leq B_P \quad (\text{discharge power}) \tag{5}$$

$$soc_{h+1} = soc_h + \eta_{ch} (x_h^{pv \rightarrow b} + x_h^{g \rightarrow b}) - \frac{1}{\eta_{dis}} (x_h^{b \rightarrow l} + x_h^{b \rightarrow g}) \quad (\text{SOC dynamics}) \tag{6}$$

$$\underline{s} B_E \leq soc_h \leq \bar{s} B_E \quad (\text{SOC bounds}) \tag{7}$$

The state of charge is cyclic: soc returns to its starting value over the full year in 8760 mode, or over each representative day in 288 mode. By default grid charging is disabled ($x_h^{g \rightarrow b} = 0$) and battery-to-grid export is permitted.

4 Coefficients and conventions

The capital-annualization factors convert an up-front capital cost into an equivalent annual cost over the analysis horizon N :

$$\alpha_{pv} = \text{CRF}(r, N), \quad \alpha_b = \frac{K_{batt}}{\text{PVA}(r, N)}, \quad (8)$$

where

$$\text{CRF}(r, n) = \frac{r(1+r)^n}{(1+r)^n - 1}, \quad \text{PVA}(r, N) = \frac{1 - (1+r)^{-N}}{r}, \quad K_{batt} = 1 + (1+r)^{-n_{batt}}. \quad (9)$$

Solar is amortized once over its N -year life. The battery term K_{batt} is the present value of *two* purchases: one at installation and a replacement at year n_{batt} within the N -year horizon. Because of that replacement, $\alpha_b > \text{CRF}(r, n_{batt})$.

Prices and rates. p_h^{exp} is the NEM 3.0 Avoided Cost Calculator export credit, supplied as a month $\times$ hour table per utility. p_h^{imp} is the retail time-of-use import price for the county's utility and rate plan.

Defaults. $r = 7\%$, $N = 25$ yr, $n_{batt} = 15$ yr, round-trip efficiency $\eta_{ch}\eta_{dis} = 0.96$, state of charge confined to 20%–90% of B_E , and $c_{bp} = c_{deg} = 0$. The 15-year battery life and battery cost follow NREL ATB 2024 Residential Battery Storage; solar cost follows the CPUC-adopted installed price.

Scope of the objective bill. The bill inside the objective is a simplified hourly accounting (imports at p_h^{imp} , exports at p_h^{exp}) used to drive the sizing and dispatch decision. The full NEM 3.0 monthly accounting (non-bypassable charges, fixed charges, minimum bill, carry-forward credits, and annual true-up) is applied downstream (Step 12) when computing the reported bill.

5 Incentive policy regime

The capital costs c_{pv} and c_b entering the objective are *net* of whichever incentives legally exist in the modeled installation year. Which incentives exist is a policy question separate from how much of an available incentive a given household actually captures, and the model treats the two as orthogonal axes: a *policy regime* determines the set of live incentives, and an *incentive scenario* (full / half / none) scales what is captured.

Two regimes are defined. Under ITC_2025, the federal residential clean energy credit (IRC 25D) supplies 30% of installed cost, giving $c_{pv} = \$2,310/\text{kW}$ and $c_b = \$1,022/\text{kWh}$. Under POST_ITC_2026, IRC 25D is repealed and the gross costs apply directly: $c_{pv} = \$3,300/\text{kW}$ and $c_b = \$1,461/\text{kWh}$. The One Big Beautiful Bill Act (Pub. L. 119-21, 2025-07-04) repealed IRC 25D for expenditures made after 2025-12-31, so POST_ITC_2026 is current law and is the default regime for this analysis. The ITC era is retained as an explicit comparison regime rather than as the headline case.

Federal and state incentives in the appliance layer

Table 1 lists every incentive the cost model applies, the call site that applies it, and its status under OBBBA. The table is maintained in code as `INCENTIVE_REGISTRY` in `appliances/incentive_policy.py`; paths in the first column are relative to `appliances/` and line numbers are as of 2026-07-17.

Call site	Credit	Statute / program	Status
<code>solar_system.py:27</code>	30%	IRC 25D	Repealed after 2025-12-31
<code>battery_storage.py:27</code>	30%	IRC 25D	Repealed after 2025-12-31
<code>electric_heating.py:45</code>	30%, cap \$2,000	IRC 25C	Repealed after 2025-12-31
<code>electric_water_heating.py:46</code>	30%, cap \$2,000	IRC 25C	Repealed after 2025-12-31
<code>electric_vehicle.py:37</code>	\$7,500	IRC 30D	Terminated after 2025-09-30
<code>electric_cooking.py:29</code>	\$840	IRA §50122 (HEEHRA)	Not repealed [†]
<code>electric_vehicle.py:56</code>	\$3,000	CA utility rebate	Not repealed [†]
(not applied)	\$200/kWh	CA SGIP general market	Closed to new applicants

Table 1: Incentives applied by the capital-cost model, with status under the One Big Beautiful Bill Act (Pub. L. 119-21). [†]HEEHRA and the California utility vehicle rebate are not federal tax credits and were not repealed by OBBBA; their availability is a program-funding question rather than a statutory one, and remains to be confirmed.

Direction of effect. The repeals do not all push the same way. Removing IRC 25D raises the cost of solar and storage, which makes storage *less* likely to enter the optimal solution and therefore reinforces the finding that storage does not pencil out under NEM 3.0. Removing IRC 25C and IRC 30D raises the cost of the heat pump, the heat pump water heater, and the vehicle, which makes full electrification more expensive relative to a gas baseline and therefore cuts *against* the finding that electrification is cost-competitive. The two effects are separated in the results by applying the repeals in sequence rather than jointly, so that each change in outcome is attributable to a specific policy change.

Regime boundary. The two-regime formulation is an approximation at the boundary. IRC 30D terminated on 2025-09-30, roughly three months before IRC 25C and IRC 25D, so no single date cleanly separates the regimes. For a 2026-vintage analysis, in which all three credits have lapsed, the approximation has no effect on results.